Video In Policy Out (VIPO): From Any Egocentric Video to a Dexterous Robot Policy

V2D Challenge 2026 — Track 3 Technical Report

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Abstract: Dexterous manipulation is limited by the cost of collecting robot demonstrations. Egocentric human videos offer a scalable alternative, but converting them into physically plausible robot behaviors remains challenging. We introduce **Video In Policy Out (VIPO)**, a framework that transforms a single egocentric stereo video into a dexterous manipulation policy without requiring robot demonstrations, object scans, or motion capture. VIPO reconstructs 4D hand–object interactions through whole-sequence pose selection and physics-aware refinement, then retargets the recovered motions to a robotic hand for reinforcement learning in simulation. On V2D Track 3, VIPO achieves an AUC of 0.584 and a mean position error of 6.06 cm using only its own reconstructions.

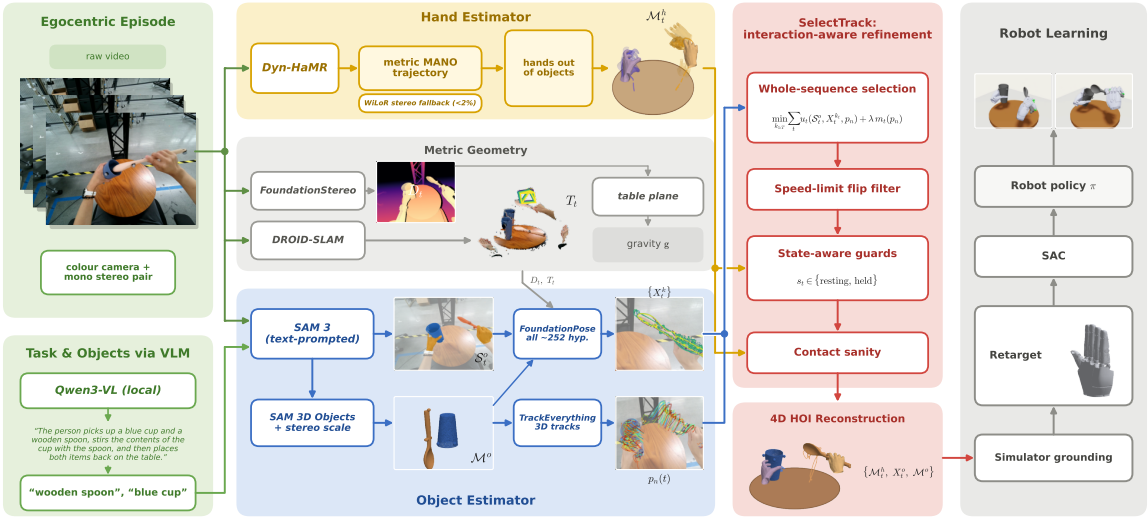


Figure 1: VIPO architecture: perception (left, centre), SelectTrack merging all evidence into one 4D hand–object reconstruction (red), and robot learning (right).

1. Introduction

Human video is a cheap source of robot skills, but it must first become a *physically imitable* demonstration: metric meshes, object trajectories and hands in one world frame, consistent with contact and gravity. In V2D Track 3 [9] a policy for floating Sharpa hands must re-enact an egocentric recording in the FlashCHORD simulator [24]; the objects it moves are scored against hidden ground truth. Chaining off-the-shelf estimators is not enough: at the hand–object interface occluded or near-symmetric objects flip by up to 180°, held objects spin in the hand, resting objects tilt into the table and fingers sit inside cups—targets no robot can imitate.

VIPO (Fig. 1) reconstructs the 4D hand–object interaction from the video with open foundation models, refines it until it is physically consistent, grounds it in simulation and learns a robot policy from it. Contributions: (i) a complete pipeline from one stereo video to a dexterous policy, with no robot demonstrations, scans or mocap; (ii) interaction-aware refinement of hand and object poses and simulator grounding, which turn a visual reconstruction into a demonstration a robot can imitate; (iii) a modular, largely automated design (scripted stages; the few manual inputs are listed in App. A).

2. Approach

VIPO runs in four stages (Fig. 1): perception estimates geometry, hands and object-pose hypotheses; SelectTrack refines them into one consistent hand–object trajectory; simulator grounding makes the scene physically stable; and reinforcement learning trains the robot policy. All estimates live in one metric world frame whose z-axis is the table normal, and

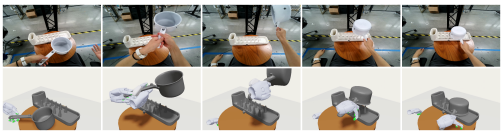


Figure 2: Human (top) vs. our Sharpa policy (bottom), e28 pot into dish rack (*success*). More successes and failures: App. Figs. 3–4.

the stages exchange files, so each estimator can be replaced on its own.

Perception. FoundationStereo [21] provides metric depth D_t and DROID-SLAM [16] the head-camera poses T_t; together with a fitted table plane they define the world frame. Qwen3-VL [1] names the task and objects, and these names prompt SAM 3 [2] for object and hand masks. SAM 3D Objects [12] builds a metric mesh per object from unoccluded views, and DynHaMR [22] recovers MANO [11] hands (98% of frames, with a WiLoR [10] stereo fallback). For objects we keep *all* ∼252 FoundationPose [20] hypotheses X_t^k with scores s_t^k per frame, and TrackEverything [5] provides 3D point tracks $p_n(t)$ on the object surfaces.

Interaction-aware refinement (SelectTrack). This stage merges masks, hypotheses, point tracks and hands into one trajectory per object in four steps. First, a Viterbi search selects one hypothesis per frame (*L1*) by minimising $\sum_t [1 - s_t^{k_t} \mathrm{IoU}_t^{k_t} + \lambda_p \bar{d}_{\mathrm{surf}}(X_t^{k_t}) + \lambda_m \,\mathrm{mean}_n\, \bar{d}(\Delta_t p_n(t-1), p_n(t))]$, where bars denote truncated, normalised distances (weights in App. A). The pose must match the mask with hand pixels ignored, lie on the tracked points, and its frame-to-frame motion $\Delta_t = X_t^{k_t}(X_{t-1}^{k_{t-1}})^{-1}$ must move the points as they really moved; a 180° flip can fit a silhouette but not this motion. Second, a flip filter removes jumps faster than 20° or 6 cm per frame (*L2*). Third, the hands decide the object state (*L3*): an untouched object stays still, a held object rotates with its hand, and symmetric cups take the minimal twist. Finally, contact checks place resting objects on the table and push the hands out of object meshes (*L4*); related work: App. A.1.

Simulator grounding (L5). Each scene is settled for 2 s in simulation without the robot, and the settled poses become the start state. Collision meshes are CoACD [19] pieces of 4 mm wall shells, so cups and racks keep their openings.

Robot learning. The organiser converter retargets MANO to the Sharpa hand. One FlashSAC [4, 24] policy per episode tracks the object trajectory with a *contact-aware* reward that places the fingers at the human contact locations and rewards contacts able to hold the object. Up to five runs per episode (two seeds, two friction settings) are trained, and the best is kept automatically (App. A).

Submission (runs selected from)	AUC↑	SP-SR↑	MP-SR↑	RPE↓	MPPE↓	CD-O↓
organiser scans (non-compliant)	0.529	0.40	0.15	8.50	6.84	0.088
t3 (L1–L2 only)	0.504	0.50	0.15	9.23	8.60	0.407
best1 (t3, t6: + L3–L4, carving)	0.532	0.55	**0.20**	8.10	7.82	0.407
best2 (+ t6 seed 7)	0.553	0.60	0.15	7.54	7.18	0.401
best3 (+ t10, t10 fr. 1.5)	**0.584**	**0.65**	0.15	**7.35**	**6.06**	**0.401**
leader (richchentw)	0.985	1.00	1.00	0.64	0.61	0.212
our rank / teams	18/44	19/43	17/42	19/42	17/41	23/41

Table 1: Official scores on the 20 hidden test episodes (Kaggle). AUC (of ADD), SP-SR/MP-SR (loose/strict success rate), RPE/MPPE (pose errors, cm) score our policy rollouts' object poses; CD-O (cm) our meshes. Rows are a cumulative *submission progression* (per-episode best-of), not a controlled ablation. Grey: organiser scans (non-compliant).

3. Experiments & Results

What we changed and what it gained (Table 1). Each submission adds runs to the previous best-of, so gains mix components with new seeds and settings. **Policy learner** (internal, own-reference evaluation on 40 episodes): on identical data and reward FlashSAC beat PPO [13] (AUC 0.63 vs. 0.37, App. Fig. 5). **Selection (L1–L2)**: the base system t3 removed the 140–180° flips that broke simulation and reached AUC 0.504. **Guards and contact sanity (L3–L4)**, hand depenetration and dish-rack carving: AUC 0.532, MPPE 8.60→7.82 cm and our best MP-SR (0.20). **A second training seed**: AUC 0.553, MPPE 7.18 cm. **Simulator grounding (L5)** with new runs at friction 1.0 and 1.5: the largest step of the progression, AUC 0.584, MPPE 6.06 cm. More directly, resting start rotation fell 12.3° →5.0° and e38 went from unsolved to solved (own-ref. MP-SR 0→0.98) with the same object poses.

Why it fails. The policies imitate our reconstructions well: 14/20 reach MP-SR ≥0.83 against our own reference, yet only 3/20 pass on the hidden ground truth. This gap points to *object and hand pose estimation* as the dominant bottleneck, though contact dynamics and embodiment differences may also contribute. Depth leaves a ∼12° tilt and cm-level offsets in object poses, e10 has a wrong cup orientation, e32 a mis-tracked cup, the e20 pitcher mesh floated, and the e03 hand never closes on the pot. The leader's 10× lower MPPE agrees.

4. Conclusion

One egocentric stereo video and open models are enough for a dexterous robot policy, with no scans, mocap or labels, through largely automated and replaceable stages. What mattered most was making the reconstruction *physically consistent* through interaction-aware pose refinement and simulator grounding; the simulator also flagged reconstruction errors.

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A. System Details

Data flow and interfaces. Every module is a separate Slurm job (heavy stages in five Apptainer images converted from the organiser Dockerfiles: `agile`, `v2d_foundation_pose`, `trackeverything`, `flash_chord`, `rg`) that reads and writes files, so any stage can be replaced independently.

Module	Input → output	Why it exists
Calibration fix-up	published extrinsics → corrected colour→stereo-left	the published extrinsic referred to the other stereo camera (18.9 px → 3.4 px)
FoundationStereo	stereo pair → metric depth D_t in the colour camera	metric scale for everything downstream
DROID-SLAM	video, stereo scale → camera poses T_t	one world frame for hands, objects and tracks under head motion
Table fit	depth → plane + disk (r=0.2375 m)	gravity = table normal; support surface for L4/L5
Qwen3-VL	8 frames → task text, one noun phrase per object	task-agnostic object vocabulary for segmentation
SAM 3	video + phrases → object and hand masks $\mathcal{S}_t$	silhouette evidence, hand-pixel exclusion, track seeding
SAM 3D Objects	unoccluded views + depth → metric mesh $\mathcal{M}^o$	own meshes for CD-O, pose registration and simulation
Dyn-HaMR / WiLoR	video, stereo → MANO trajectory $\mathcal{M}_t^h$	human hand motion and contacts for retargeting and reward
FoundationPose	D_t, $\mathcal{S}_t$, $\mathcal{M}^o$ → all ∼252 hypotheses X_t^k, scores s_t^k	keep the right answer even when it is not the argmax
TrackEverything	D_t, T_t, masks → 3D tracks $p_n(t)$	motion evidence that disambiguates flips and spins
SelectTrack L1–L4	all of the above → object trajectories, depenetrated hands	a physically imitable 4D reconstruction
Export + grounding (L5)	reconstruction → FlashCHORD scene	table at z=0, settled starts, collision assets
Retarget + FlashSAC	scene, MANO → Sharpa reference, policy	robot re-enactment, scored rollouts

SelectTrack layers (all scripted, no learned critic). L1 selection: unary = 1−score×silhouette IoU (hand pixels excluded) + tracked-point distance to the candidate surface; pairwise = how well $X_j X_i^{-1}$ reproduces the tracked points' motion, plus weak, symmetry-aware smoothness; then a long-range refinement anchoring every track in the object frame (points seen at frames 50 and 200 must land on the same mesh spot). Tracks are seeded every 5 frames on SAM 3 masks minus dilated hand pixels and depth edges; for tractability, subsets of 128 (unary), 64 (pairwise) and 200 (refinement) tracked points are used (fixed seed). Weights (`stereo/track_pose.py`): score normalised per frame; point term λ_p=1, truncated at 2 cm; motion term λ_m=2, truncated at 3 cm; smoothness $0.1\times(|\Delta p|/2\,\text{cm} + |\Delta R|/0.3\,\text{rad}$, minimum over the symmetry group), raised to 0.5 when fewer than 6 tracked points are shared; refinement uses a Huber loss (1 cm) and a pose prior of 0.15 rad / 2 cm. Viterbi is exact, $O(TK^2)$ with $K \approx 252$: the full $K\times K$ transition cost is computed per frame in batches of 32 candidates; frames without candidates are interpolated. **L2 pose filter**: symmetry unwrap; Viterbi over keep / flip (about one of 3 principal axes through the centroid) / skip with hard limits of 20° and 6 cm per frame; slerp fill; light smoothing. **L3 guards**: rest hold (untouched objects do not move); spin guard (held: rotation follows the holding hand + ≤3°/frame slip; not held: ≤1°/frame); twist unwrap (exact minimal twist of rotationally symmetric cups about their axis—a snapped twist was noise that the non-symmetric simulator reward made the robot chase). **L4 contact sanity**: untouched objects near the table sit exactly on it (stacked objects skipped); each hand is translated (time-smoothed) out of every object, cups treated as solids, because reference fingers 2–4 cm inside cups made the simulator shove hand and object apart at reset. **L5 grounding**: world rotated so the table normal is $+z$; rest settle onto a stable hull face; table depenetration; simulator settling (2 s without the robot, blended over 5 frames, accepted only with ≥5 rest frames, ≤20° and ≤3 cm; mean start rotation 21° →5°); realistic masses (cups 0.08 kg, pot 0.45 kg); CoACD pieces of a 4 mm wall shell (openings such as cup mouths and rack slots stay open); dish-rack pieces removed where another object really sat; solid cup bases.

A.1. Relation to prior refinement work

Hand2Robot [3] performs object-centric 4D hand reconstruction for robotics with multi-hypothesis selection; our L1 keeps every FoundationPose hypothesis in the same spirit but chooses among them over the whole sequence, with TrackEverything point tracks (TAPIP3D [23] is the related persistent-3D tracker and could supply the same unary and pairwise evidence) providing the motion term. VICON [6] fuses visual, inertial and contact cues in a factor graph and ComPose [15] learns when to trust the hand for object tracking; our L3 guards express the same coupling as explicit rules (held objects follow the hand, untouched objects stay still). We implemented a VICON-style soft hand–object pose graph: it is accurate on average but, like other local refiners (Diff-DOPE [17], GoTrack [8]), cannot leave a 140–180° flip basin. GeneOH Diffusion [7] denoises hands given fixed objects and cannot repair object poses; refitting MANO to its joints only twisted the fingers. An AGILE-inspired [14] agentic critic-and-corrector loop was dropped because it was performing poorly. DeepSimHO [18] motivates our simulator-in-the-loop direction; our current L5 uses the simulator only to settle resting starts.

Meshes. Several SAM 3D samples per object; the one with minimum median Sim(3)-registered Chamfer distance to the others is kept. The wooden spoon (no public scan exists) is built the same way. The water pitcher is stretched 27.4→31.0 cm at its base (it floated 36 mm above the table in all three pitcher episodes); submitted for e20 only.

Policy learning. FlashSAC, one policy per episode, 2048 parallel worlds, 250 M environment steps (3–7 h on one NVIDIA L40S), real-time playback at the video rate (20.254 Hz). Termination limits start at 3× the official ones and tighten to 10 cm / 0.5 rad; assistance is annealed to 0. Contact friction 1.0 or 1.5. FlashSAC beat PPO [13] on identical data and reward (AUC 0.63 vs. 0.37 on 40 episodes; median progress 0.62 at 250 M steps vs. 0.27 at 492 M, Fig. 5). Runs are selected by own-reference MP-SR, then SP-SR, then AUC with the official (deterministic) evaluator. The reward is the same in every run:

Term	Weight	Purpose
object keypoints	40	object pose tracking
contact location / contact location touch	5 / 2	robot fingers where the human's were
contact wrench support	10	contacts able to carry the object
missed / unintended contact / unintended contact location	−1 / −10 / −1	contact consistency
hand keypoints / hand joint positions	0.25 / 0.25	weak hand imitation (observed frames only)
reference progress	5 per step	all objects within 3 cm / 0.3 rad, scaled by progress
task completion	200 once	object ends at its final reference pose
termination	−100	

What each submitted episode contains (best3). Up to five training runs exist per scored episode (t3, t6, t6 seed 7, t10, t10 friction 1.5; for e20 the seed-7 run is t7, which uses the lengthened pitcher); the submitted one is chosen per episode as above.

Run (episodes)	Object poses	Hands	Simulation assets	Friction
t3 (e01 e14 e15)	L1 + L2	Dyn-HaMR / WiLoR	4 mm walls, no carving	1.0
t6, t6 seed 7 (e03 e32 / e06 e10 e30 e35 e37)	L1–L4	+ pushed out of objects	+ rack carving	1.0
t7 seed 7 (e20)	L1–L4	as t6	+ pitcher lengthened	1.0
t10 (e16 e28 e33 e36)	L1–L4	as t6	+ settled starts, cup bases	1.0
t10 friction (e05 e08 e19 e25 e38)	L1–L4	as t6	as t10	1.5

Manual inputs and exceptions (complete list). (1) Submitted SAM 3 masks used a hand-written prompt table (the Qwen3-VL stage now reproduces it, 35/36 objects within 0.02 IoU); (2) water-pitcher mesh lengthened, e20 only; (3) dish-rack collision pieces removed where another object sat (t6 and later). Everything else, including run selection, is scripted. **Reproduction**: the per-episode run and simulator tree of best3 are listed in `pipelines/t3/best3_selection.txt`; `pipelines/t3/pack_best.sbatch` rebuilds the submission from it.

B. Industry Innovation Award

Criterion	How VIPO addresses it *(evidence)*
Economic viability	Input: one head-mounted stereo recording—no scanning, mocap rig, markers or annotation; open models run locally on commodity L40S GPUs. The video becomes sim-ready assets (metric meshes, collision geometry, settled starts) and a reference trajectory, today's costly manual step. *(Policy training: ∼3–7 h per run on one L40S, up to five runs per episode; perception-stage cost not yet measured end to end. Own meshes vs. our organiser-scan run (different pose pipeline, so not a mesh-only comparison): AUC 0.584 vs. 0.529, SP-SR 0.65 vs. 0.40, MPPE 6.06 vs. 6.84 cm; CD-O worse.)*
Data flywheel	*Automated data generation* (not yet a closed loop): one pipeline and reward for all 20 tasks, fixed rules, no labels, best run picked automatically, so new videos become robot data. The simulator is a free critic *(it exposed a ∼12° depth tilt, fingers 2–4 cm inside cups, a pitcher floating 36 mm; own-ref. MP-SR flags failures)*. Limits: two scripted exceptions (e20 pitcher, rack carving); submitted masks used a hand-written prompt table (the VLM now reproduces it); feeding these signals back automatically is future work.
Innovation	Challenges two assumptions: (i) hand–object capture for robots needs scans or mocap—VIPO uses one stereo video; (ii) pose accuracy alone makes a demonstration imitable—physical consistency matters as much. *(Settled starts + friction solved e38, own-ref. MP-SR 0→0.98, with no pose change.)*
Complexity reduction	One capture device, no instrumentation; 13 modules (heavy ones in 5 container images)

Table 2: VIPO vs. the four Industry Innovation Award criteria; evidence in italics (deployment notes below).

C. Additional Figures

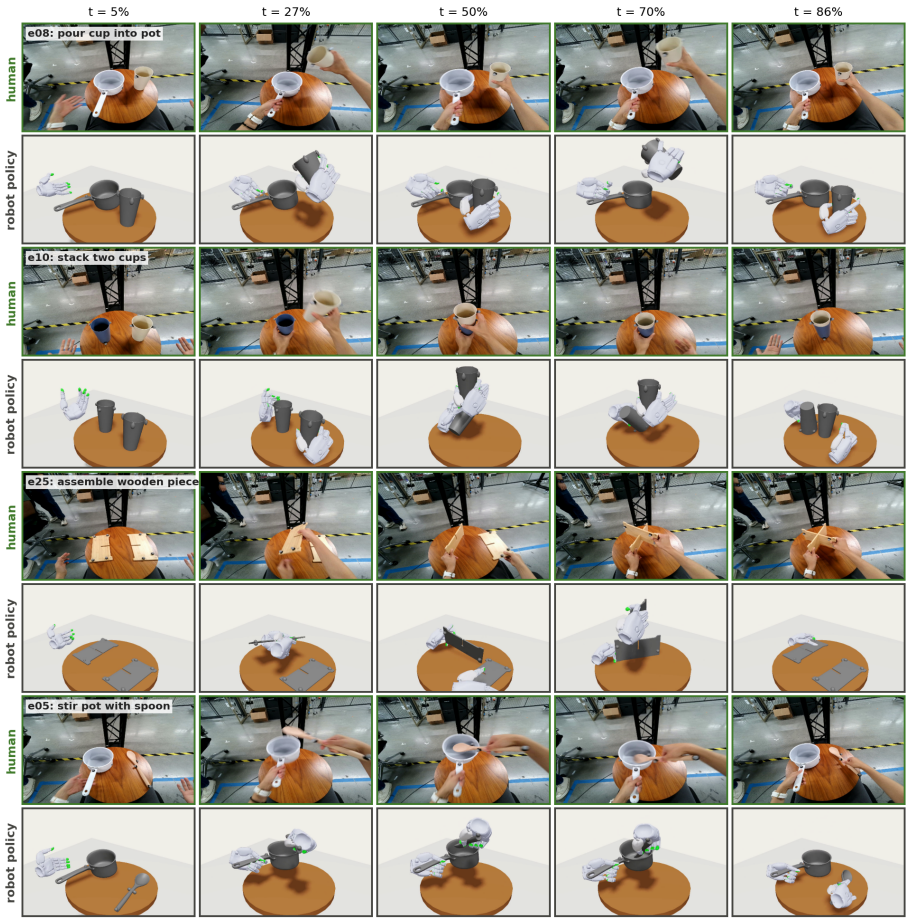


Figure 3: Human egocentric frames (green) vs. our learned Sharpa policy (grey) at matched normalised time. **e08** pour cup into pot: the cup is lifted, tilted over the pot and put back (*success*). **e10** stack two cups: the second cup tips over and the cups end side by side (*failure*; wrong cup orientation in our reconstruction). **e25** assemble wooden pieces: the policy raises one piece but lays it down again and never completes the assembly (*failure*, own-ref. MP-SR 0.49). **e05** stir pot with spoon: pot and spoon are handled throughout (*success*, 0.99).

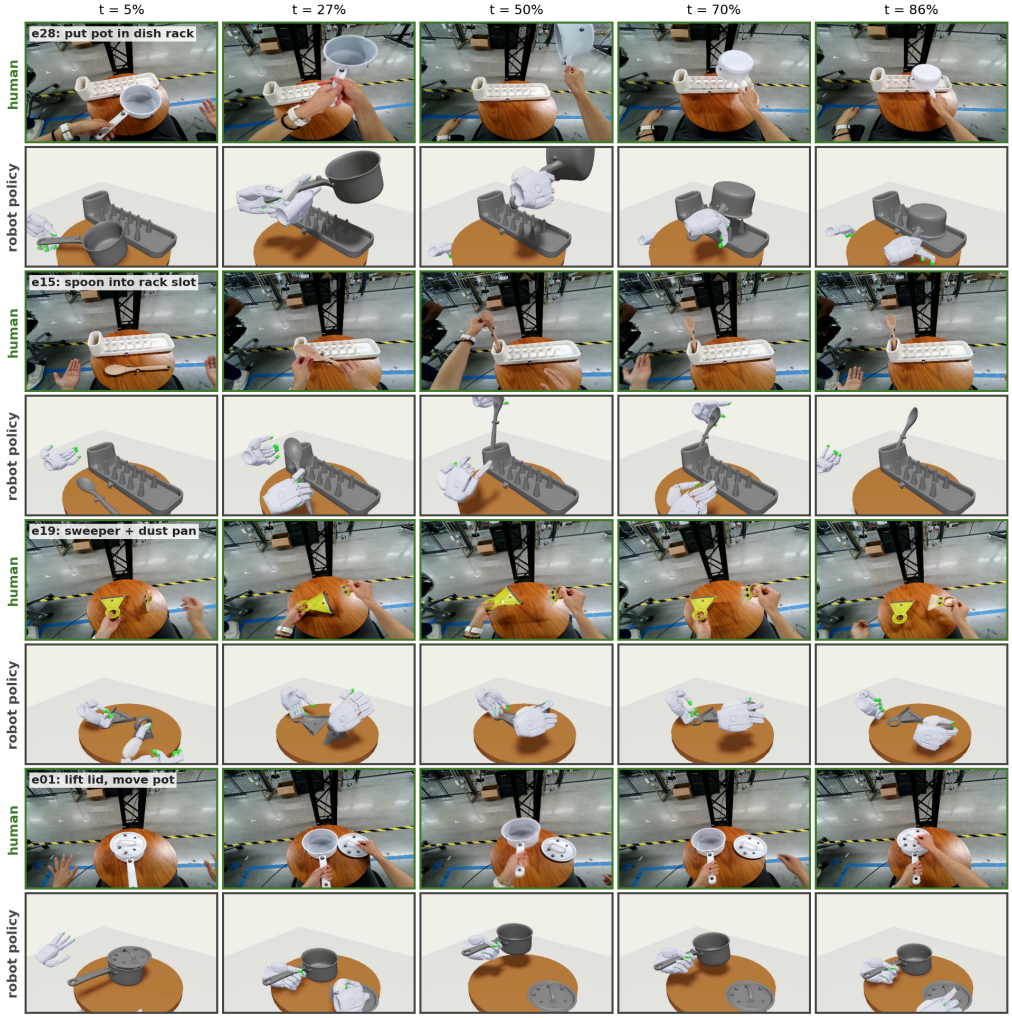


Figure 4: As Fig. 3. **e28** put pot in dish rack (*success*); **e15** spoon into rack slot: the spoon ends upright in the slot (*success*); **e19** sweeper + dust pan: two thin tools handled by both hands (*partial*, own-ref. MP-SR 0.37); **e01** lift lid, move pot (own-ref. 0.96): lid and pot are moved, but at 86% the robot's lid is still on the table while the human is putting it back on the pot (*partial*).

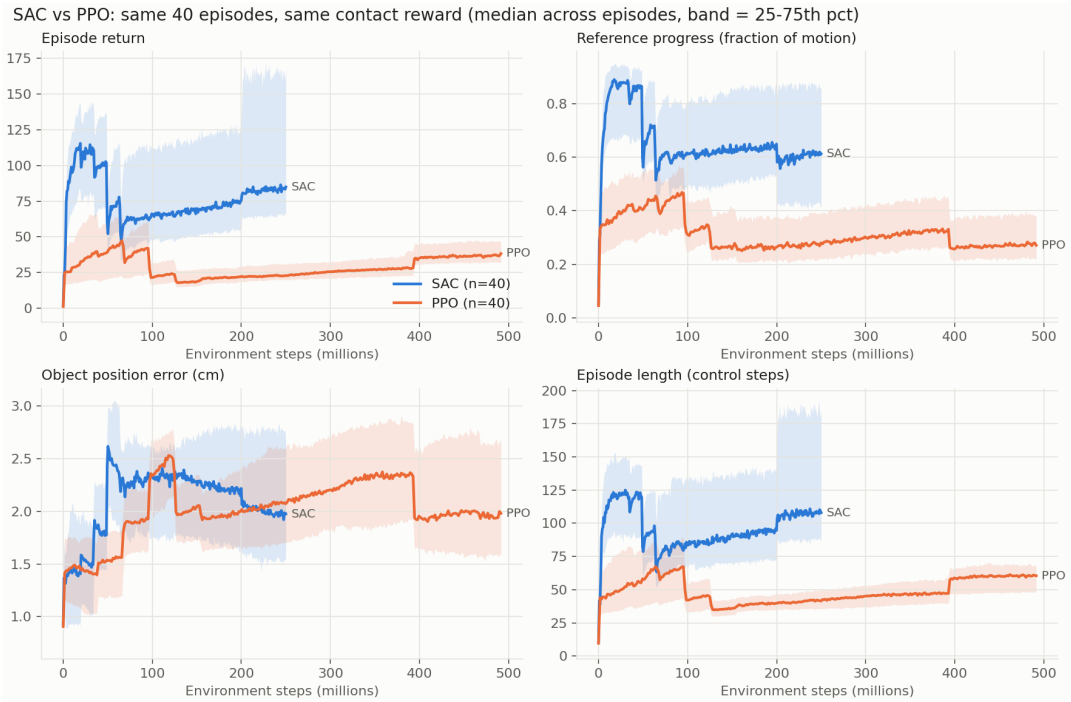


Figure 5: SAC vs. PPO on the same 40 episodes and contact reward (median, band 25–75th percentile). SAC reaches a median reference progress of 0.62 in 250 M steps, PPO 0.27 after 492 M; drops are curriculum stage changes.